



Aegis AI

Version 1.0

System Analysis Design

August of 2026

1. Problem Statement & Objectives

Problem Statement

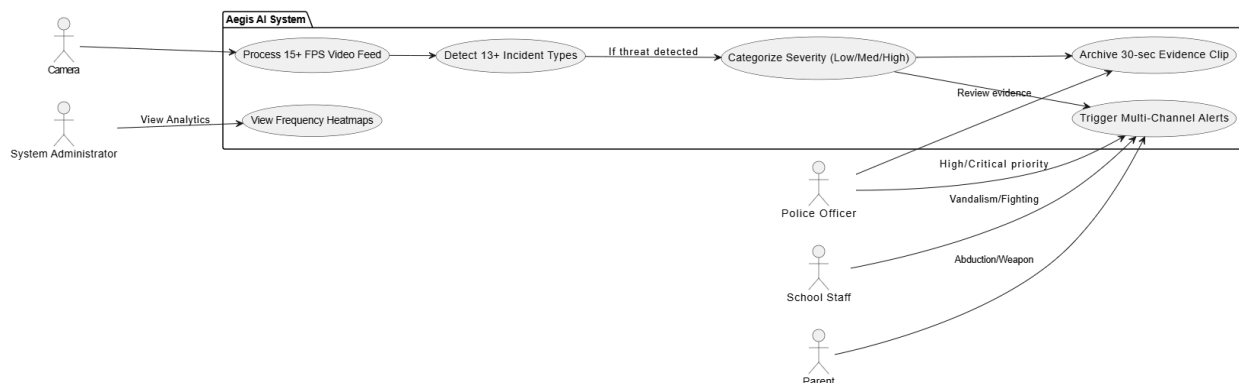
Traditional security setups require 24/7 human observation, which is prone to fatigue and often fails to prevent incidents in real time. Stakeholders including police forces, schools, and parents require a proactive system that automatically flags critical incidents (like shootings, explosions, or vandalism) to dispatch emergency units or trigger lockdowns without waiting for manual intervention.

Project Objectives

- **Active Monitoring:** Transform passive surveillance into an active system using Spatial Temporal neural networks to detect 13 specific incidents.
- **Multi Tiered Alerting:** Provide immediate, multi channel notifications (SMS, Email, Dashboard Push) to the correct stakeholders based on priority.
- **Evidence Collection:** Automatically archive 30 seconds video clips of incidents to serve as reliable evidence for legal or administrative prosecution.

Use Case Diagram & Descriptions

- **System Actors:** Security Personnel, System Administrators, Police Officers, School Staff, Students, and Parents.
- **Key Interactions:**
 - *Police:* Receive critical alerts, view live feeds and precise GPS locations, dispatch tactical units, and review 30 seconds evidence clips.
 - *Schools:* Detect fighting/vandalism, categorize severity, trigger mobile/desktop alerts, and initiate emergency protocols.
 - *Parents:* Receive emergency alerts (abduction, weapon detection, critical injury) and verify child status.



Functional Requirements (FRs)

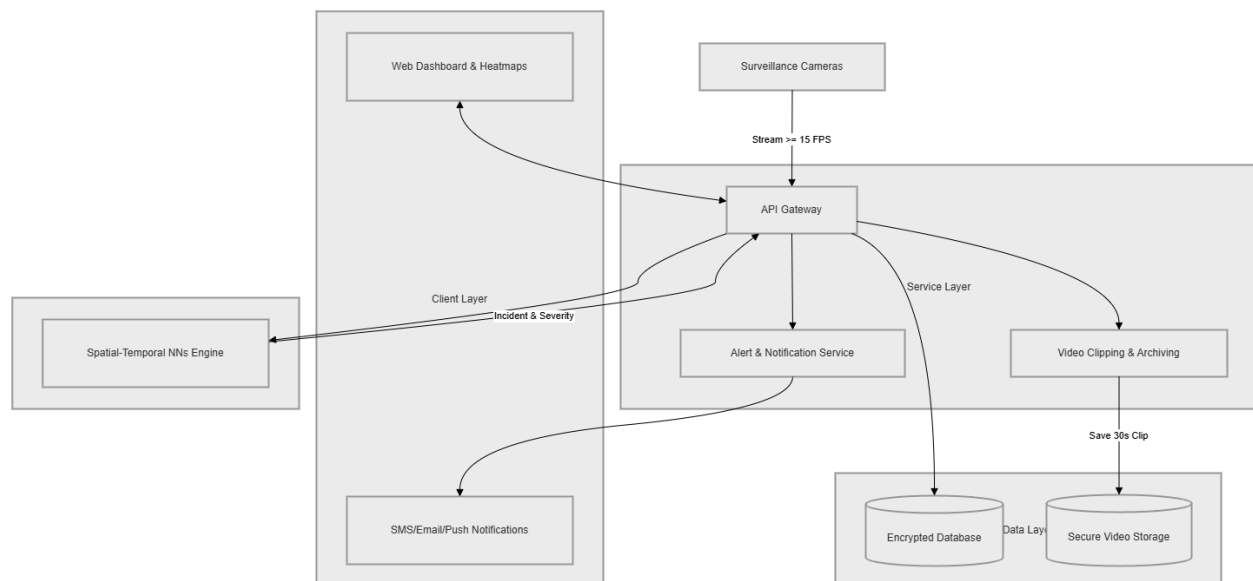
- **FR-1 (Real time Detection):** Detect 13+ specific incidents (Abuse, Arson, Robbery, etc.) within seconds.
- **FR-2 (Severity Categorization):** Automatically label incidents as Low, Medium, or High priority.
- **FR-3 (Alert Triggering):** Send notifications via SMS, Email, and Dashboard Push to designated stakeholders.
- **FR-4 (Analytics):** Provide a web based UI for live feeds, logs, and frequency heatmaps.
- **FR-5 (Archiving):** Automatically clip and store 30 seconds of video prior to and after a detected incident.

Non Functional Requirements (NFRs)

- **Performance:** Process video frames at a minimum of 15 FPS to ensure real time accuracy.
- **Security:** Encrypt all video data to prevent unauthorized access or tampering.
- **Reliability:** Maintain high uptime for life critical event detection.
- **Scalability:** Support multiple camera feeds without a drop in processing speed.

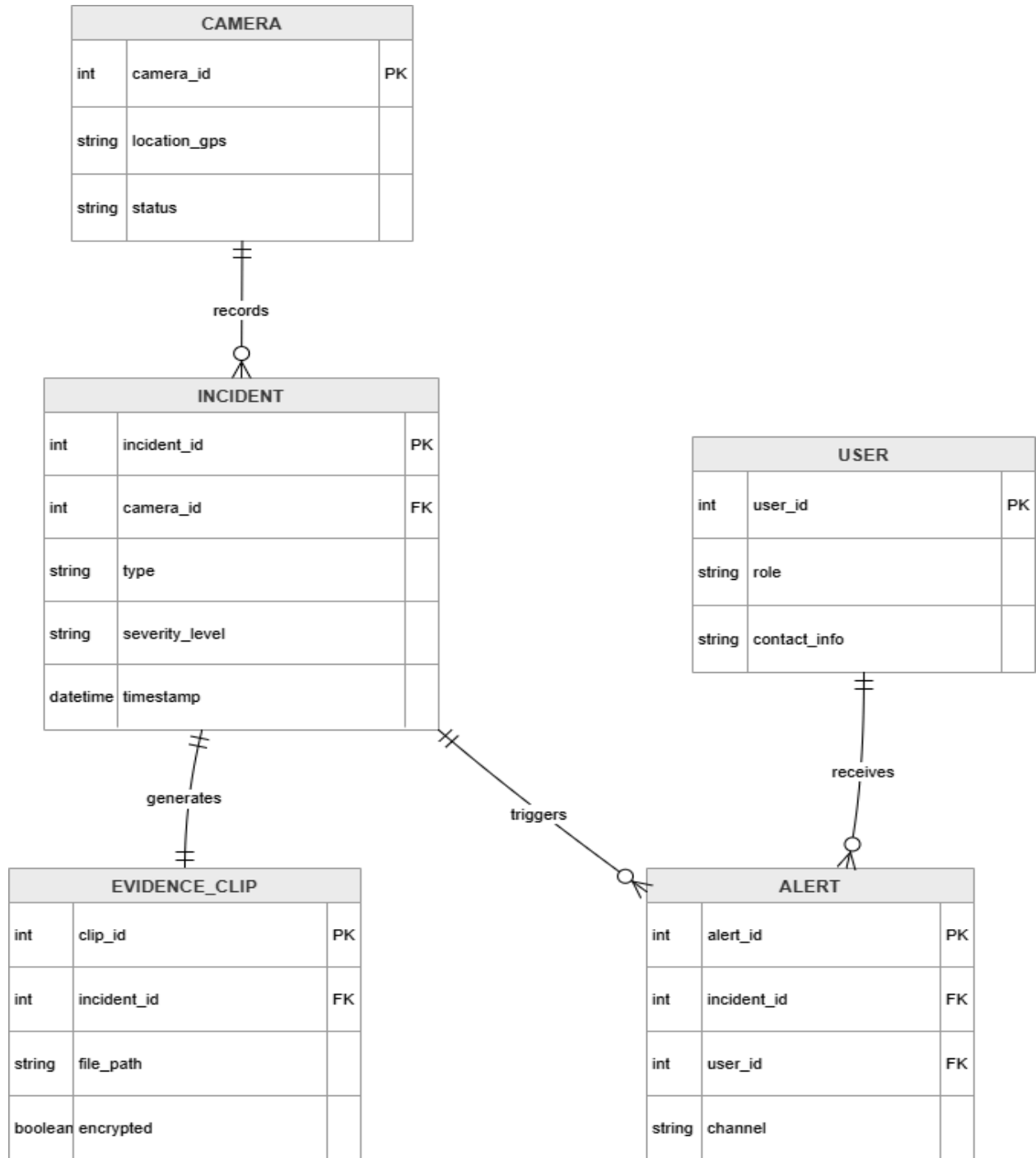
Software Architecture

The system adopts an API Driven Microservices architecture to handle the distinct loads of video streaming, AI inference, and multi channel alerting.



2. Database Design & Data Modeling

Entity Relationship Diagram (ERD)

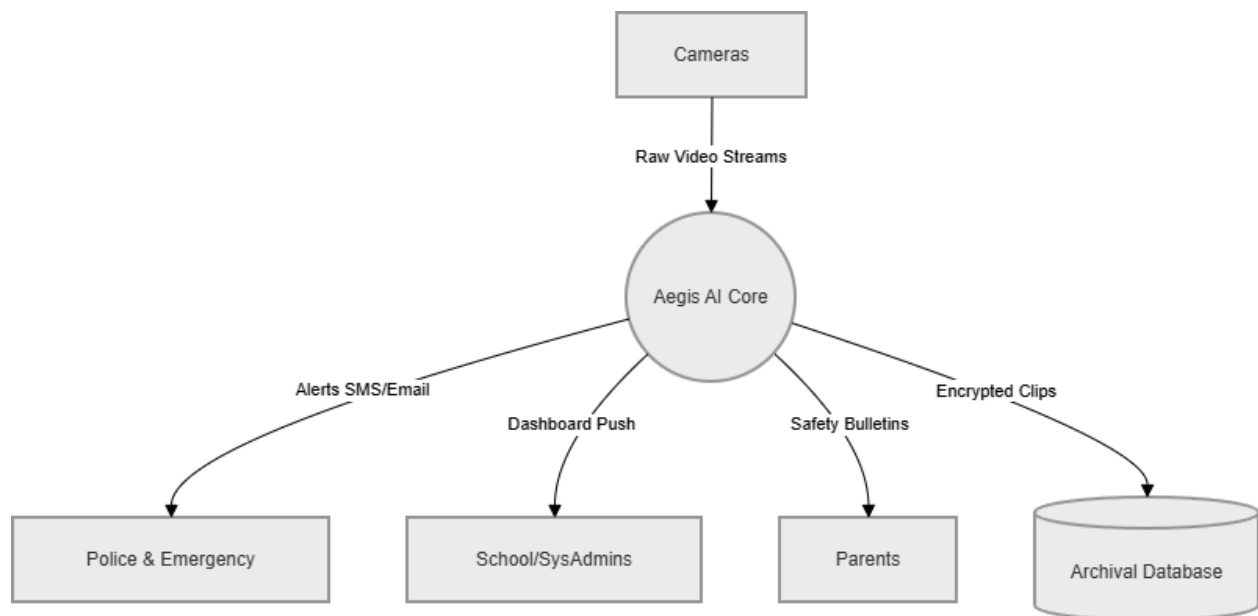


Logical & Physical Schema Considerations

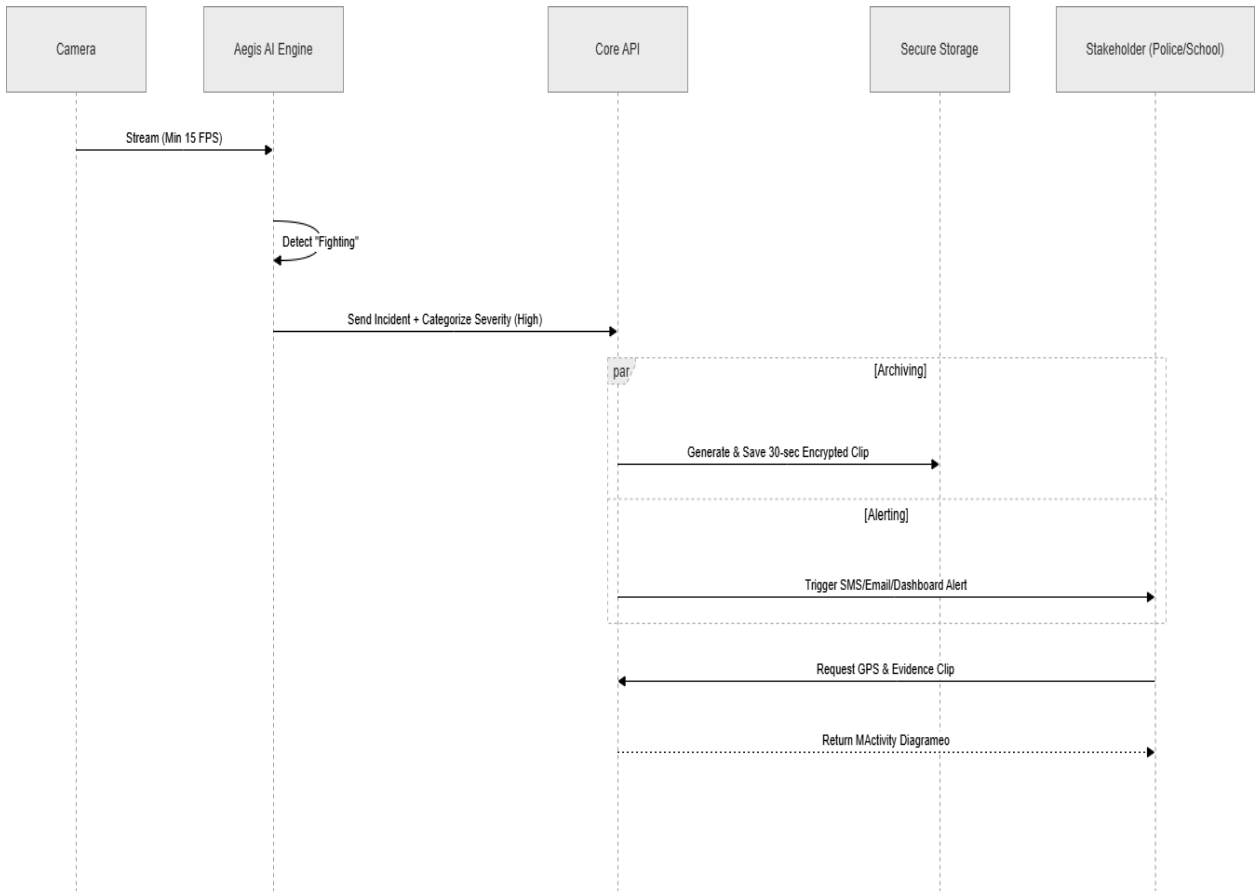
- **Normalization:** Tables are designed to 3NF to prevent data redundancy.
- **Attributes:** The INCIDENT table includes a severity_level (Low, Med, High) based on FR-2. The EVIDENCE_CLIP table includes paths to secure blob storage for the 30 second clips based on FR-5.

3. Data Flow & System Behavior

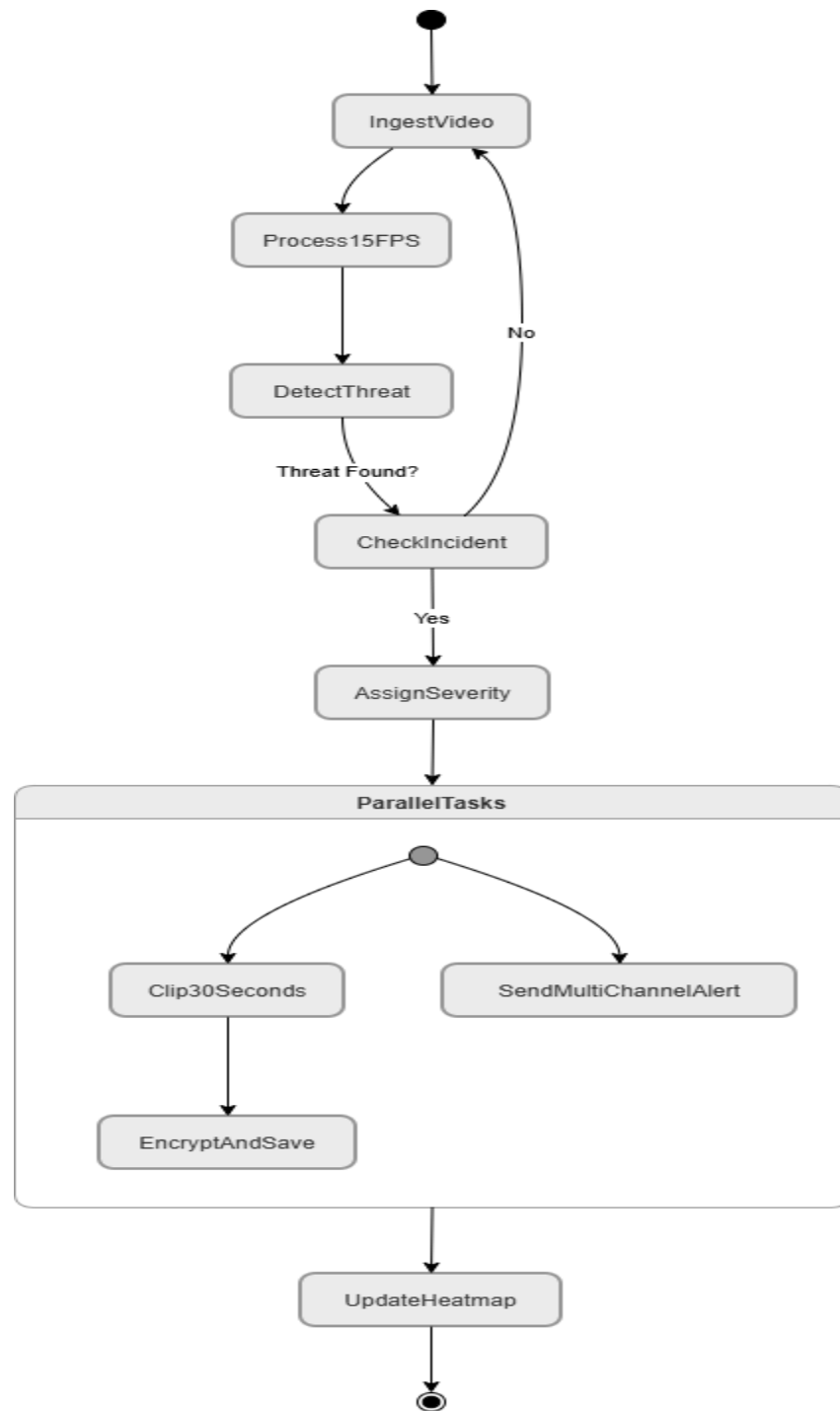
Context Level Data Flow Diagram (DFD)



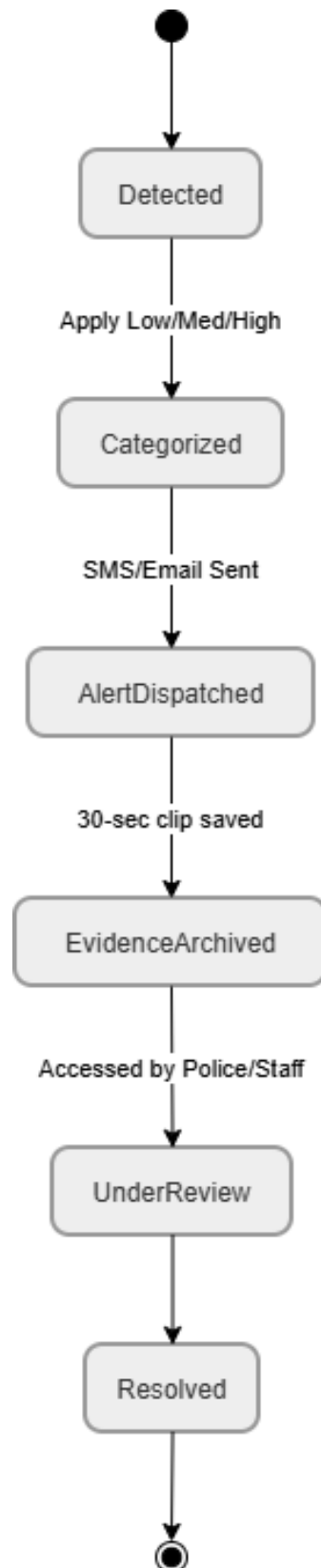
Sequence Diagram: Incident to Alert Workflow



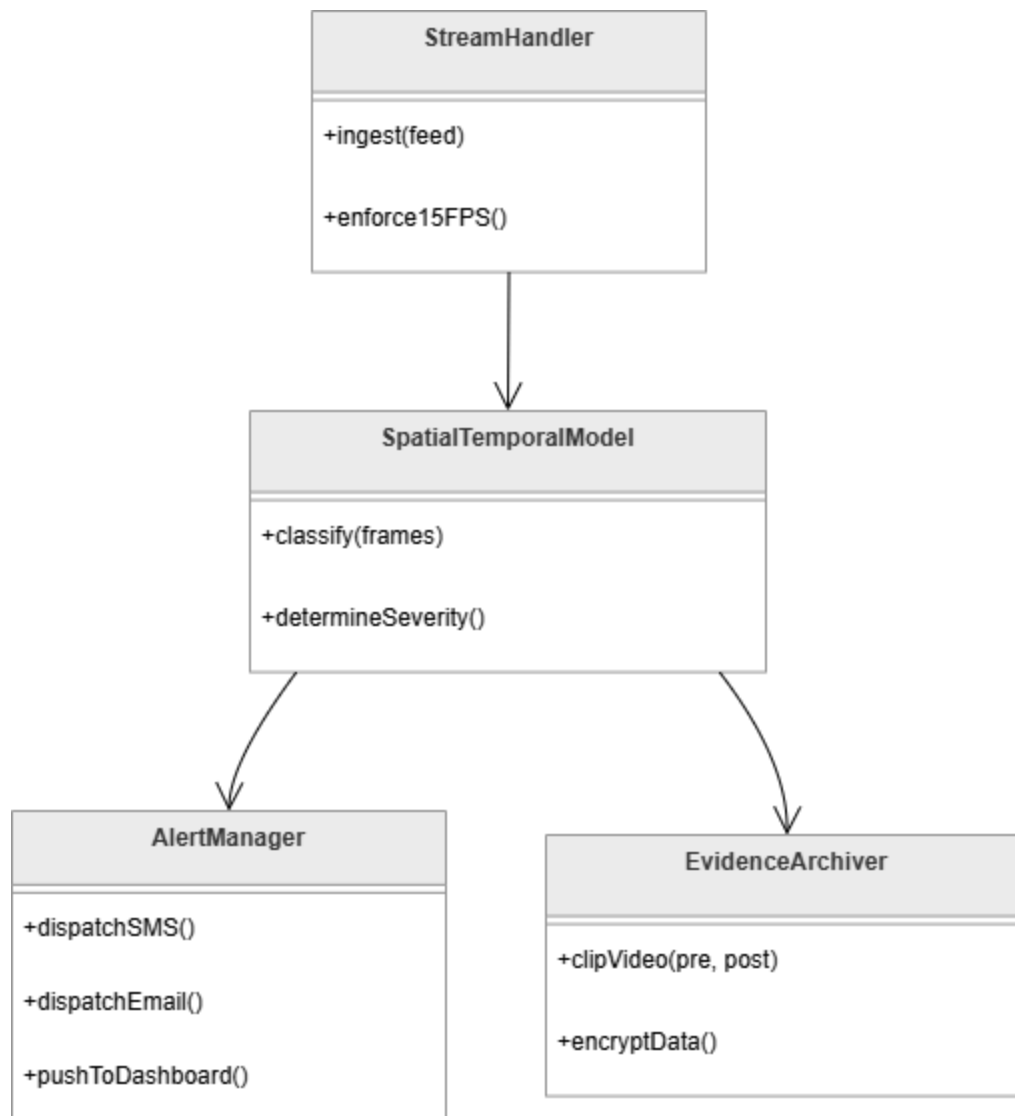
Activity Diagram



State Diagram (Incident Object)



Class Diagram



4. UI/UX Design & Prototyping

Wireframes & Mockups Strategy

- **Police/Emergency View:** A high contrast GIS/Map centric view showing precise real time incident locations. Clicking a map pin instantly plays the 30 seconds evidence clip.
- **School Principal/SysAdmin View:** A centralized grid layout of cameras with a dedicated sidebar for Frequency Heatmaps (FR-4) to identify campus danger zones.
- **Parent View (Mobile Responsive):** A simplified, reassuring interface prioritizing child status verification and critical alerts.

UI/UX Guidelines

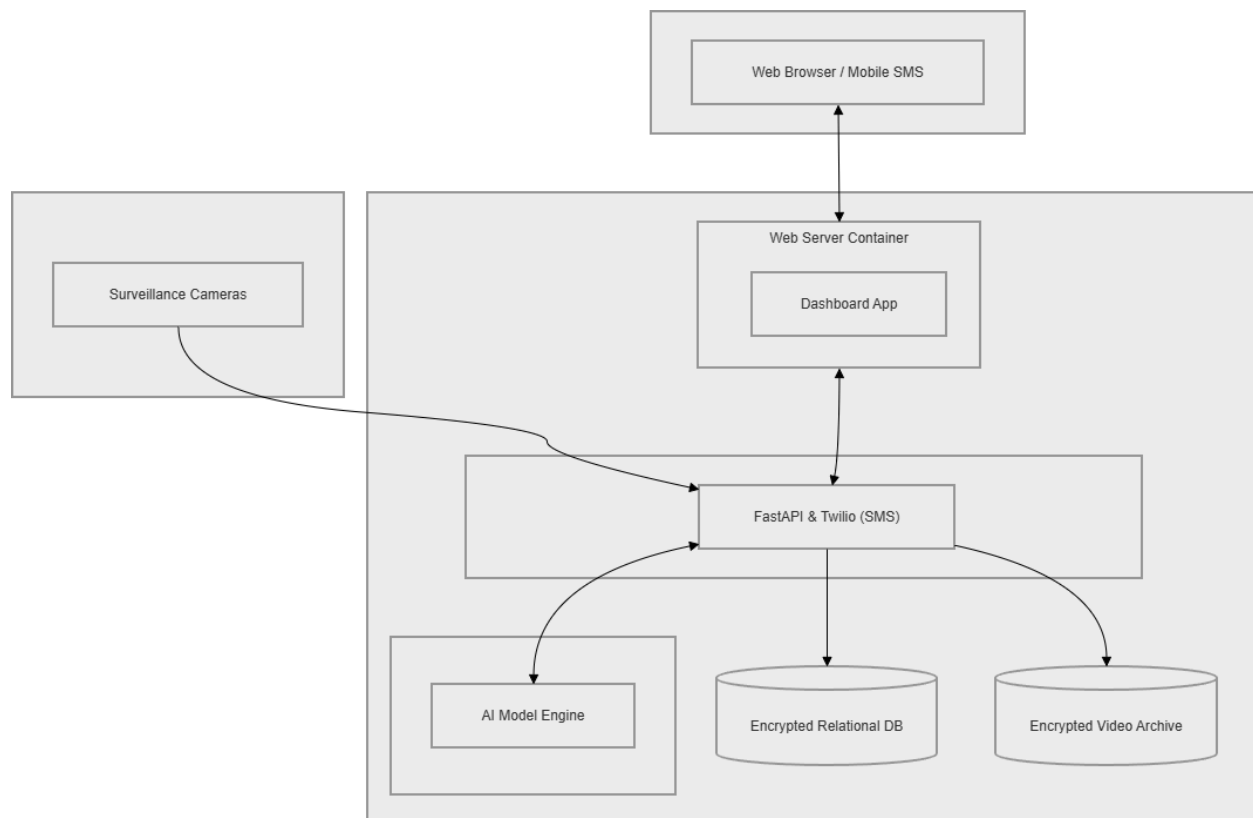
- **Color Scheme:** Neutral, dark mode palettes to reduce eye strain, reserving strict color codes for severities: Yellow (Low), Orange (Medium), Red (High/Critical).
- **Typography:** Highly legible, sans serif fonts for instant readability under stress.
- **Accessibility:** Cross platform web UI compatible with desktop control rooms and mobile web browsers.

5. System Deployment & Integration

Technology Stack

- **Backend & APIs:** FastAPI / Flask (Python) to handle async video processing and API routing.
- **Frontend:** React.js or Vue.js for dynamic dashboard rendering and heatmaps.
- **AI Models:** PyTorch/TensorFlow utilizing Spatial Temporal NNs.
- **Database & Storage:** PostgreSQL for relational data, Azure Storage / local encrypted blobs for the 30 seconds video clips.

Deployment Diagram



6. Additional Deliverables & Grading Criteria

- **Real time Performance Testing:** Validate that the system reliably ingests feeds at a minimum of 15 FPS without bottlenecks.
- **Model Accuracy:** Run evaluation metrics against the test set to ensure high precision across the 13 defined incident types.
- **Integration Testing:** Ensure the API seamlessly connects the AI predictions, the automated 30 seconds clipping, and the alerting services.

Future Integration Vision (Literature Review Additions)

While focusing on the current scope, the architecture is designed to accommodate suggested improvements in future iterations:

- **GANs / IR Imaging** for advanced low light detection.
- **Audio Integration** for multi modal alerting (gunshots/screams).
- **Dedicated Mobile Application** for field response ecosystems.

Thanks DEPI